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The design and evaluation of a visual narrated storytelling concept to improve end users understanding of explanations from a conceptual AI assistant

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ABSTRACT

Explainable Artificial Intelligence methods and their presentation tend to dive deep into the technical details of complex models, making them difficult for domain users to understand. A particular challenge lies in enabling new users to build confidence and understanding of a digital assistant. We developed a visual narrated storytelling explainer concept and prototype to support a human operator's understanding of a conceptual AI-based digital assistant's decision-making process in managing an emergency in a future Urban Air Mobility traffic management scenario. The design exploration focused on how visual narrated storytelling can be applied to effectively convey the decision-making processes of a digital assistant supporting the proposed novel human role Urban Air Mobility Coordinator. The concept was designed using several dimensions: level of detail, temporal considerations, and modalities (visuals and narration). A mixed-method within-participant simulator experiment included qualitative and quantitative analysis and was conducted with eight Air Traffic Controllers. Participants' overarching experience, trust, and perceived appropriate level of detail of the storytelling concept was assessed. The storytelling concept helped users better understand how the digital assistant made decisions.

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Explainable AI; storytelling; construal level theory; urban air mobility; digital assistant; understanding

1. Introduction

Urban Air Mobility (UAM) and unmanned aerial systems (UAS) are foreseen to transform urban transportation, prompting ongoing efforts to establish a designated highly automated airspace framework and regulatory concept, referred to as U-space in Europe, designed specifically for drone and air taxi operations at low-level altitudes (Eurocontrol 2023). U-space operations represent a considerable evolution from current day Air Traffic Management (ATM) systems and Air Traffic Control (ATC) procedures, which rely heavily on human operators for tactical control and decision making. The envisioned U-Space ecosystem is expected to be highly automated and partly autonomous (i.e. drones with sense and avoid capabilities). The U-space traffic management system is expected to transition gradually from current day Air Traffic Management with increased automation and autonomy (Eurocontrol 2023), laying the groundwork for a new traffic management system that can accommodate the forecasted surge in drone traffic, vastly exceeding current Air Traffic Management capabilities. Increased automation and

autonomous systems, such as digital assistants, are envisioned as necessary to support human operators active in this operational environment.

Despite this roadmap, integrating advanced automation and AI into airspace management raises critical questions concerning the handling of non-normal situations and emergencies in U-space airspace operations (Lachter, Hobbs, and Holbrook 2021; Eurocontrol 2023). The U-space airspace is a designated airspace portion where drones will be integrated at the low-altitude level. It is recognised that human involvement will remain necessary at certain levels of the system (SESAR 2020), for instance to manage tasks that automated systems cannot fully manage. In addressing this gap, we envision the need for a new human role to manage traffic in the U-Space – the Urban Air Mobility (UAM) Coordinator – supported by an advanced automation assistant – the Digital Assistant for the UAM Coordinator (DUC) – designed to manage routine traffic management tasks (e.g. traffic authorisation, flight tracking, conformance monitoring, and flight information) and augment human capabilities in problem-solving and coordination.

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A challenge lies in ensuring effective collaboration between human operators and AI assistants. In aviation, the European Union Aviation Safety Agency (EASA) distinguishes between two forms of Human-AI Teaming: cooperation, in which the AI executes predefined tasks, and collaboration, where humans and AI jointly solve problems by pursuing shared objectives (EASA 2020; EASA 2024). Within the EASA framework, the Digital assistant for the UAM Coordinator (DUC) is classified as a Level 2B collaborative system, signifying its ability to engage in nuanced, goal-oriented teamwork. Effective teaming between humans and AI in safety-critical domains, such as aviation and military operations, has been identified to be essential for system safety and efficiency (National Academies 2022). For successful human-AI teams, human understanding of the AI team member has been argued as elemental for effective situation awareness and decision-making (Chen et al. 2014; National Academies 2022; O'Neill et al. 2022). Automation transparency and AI explainability have become key research areas, ensuring AI-based systems clarify their reasoning and outputs (Gunning et al. 2021). Research on Explainable AI (XAI) often emphasises algorithmic methods rather than the end-user perspective (Ehsan and Riedl 2020). For explanations to be effective (i.e. support understanding), it is essential to consider users' understanding requirements, expertise, and context (Miller 2019). The design of explanations comprises both how the explanation content is derived, and how and when it is presented to the user.

To address these needs, user-centered design approaches (Norman 2013) can be used to ensure that explanations satisfy end users' requirements for system understanding. In particular, cognitive psychology suggests that storytelling can serve as an effective medium for conveying complex information to enhance comprehension, retention (Bruner 1991) and emotional engagement by contextualising explanations within relatable scenarios (Segel and Heer 2010). While storytelling informs how information can be presented, it does not prescribe what should be explained (e.g. the content as derived from an XAI algorithm), when to deliver it, and whether its scope should be limited to avoid overwhelming the user in time-sensitive operational environments. Here Construal Level Theory (CLT) (McDermott and Folds 2022) provides a framework for structuring explanations, enabling information to be delivered hierarchically from broad and abstract concepts to specific actionable details. The combination of storytelling with CLT has been proposed suitable for presenting explanations that meet user needs, tailored to the situation and of suitable duration. (McDermott and Folds 2022).

This study explores the use of visual narrated storytelling, structured by CLT, to explain recommendations made by the conceptual digital assistant (DUC) for the novel human U-space traffic management role the UAM Coordinator in the context of managing urban airspace traffic (i.e. the U-space airspace). The goal was to support initial encounters between the UAM Coordinator and DUC, specifically by clarifying the decision-making processes involved in emergency rerouting scenarios. The format leverages both engaging visual narratives and expert-informed explanation content, presenting it in a layered manner as guided by CLT.

To evaluate this approach, a visual narrated storytelling explainer was designed and assessed in a low-fidelity simulator experiment with eight Air Traffic Controllers (ATCOs), acting as UAM Coordinators responsible for managing drone traffic in a futuristic Stockholm U-space airspace. This role, although conceptual, (proposed by the authors in [Carl Westin et al. 2025]) is expected to evolve from current Air Traffic Management and Air Traffic Control practices, making Air Traffic Controllers (ATCOs) key participants in the early design phase and concept evaluations. Thus, while the participants were qualified ATCOs, they were first-time users of the designed system (DUC and the storytelling explainer).

The study aimed to:

- Explore how a storytelling concept using visuals and narrations can be designed to convey system explanations in critical decision-making contexts.
- Evaluate in a real time decision-making context how explanations presented through visual narrated storytelling affect users understanding and attitudes toward a conceptual digital assistant and its explanations.

The work presented offers empirical insights into the design and evaluation of a storytelling-based explainer. The findings suggest that visual narrated storytelling, structured according to CLT, is a promising format for improving explainability and user comprehension in complex, time-pressured contexts, while also highlighting the need for greater temporal flexibility and interactivity to support diverse information needs.

This paper is structured as follows: Section 2 introduces the domain of Urban Air Mobility (UAM) and U-space traffic management and our proposed UAM concept as a use case. Section 3 presents the theoretical background regarding human AI teaming, Explainable AI (XAI), Construal Level Theory (CLT), and storytelling as an approach for presenting explanations. Section 4 describes the method used divided into subsections of

the scenario, working position, design process and simulator experiment. Section 5 presents results from both questionnaire responses and debriefings with participants. In Section 6, we discuss the findings and reflect on the methods used. Section 7 concludes the paper.

2. The use case: the UAM coordinator and U-space traffic management

In Europe, the CORUS research umbrella (Eurocontrol 2019) has led the work towards defining and implementing U-space operations. Using the CORUS-XUAM concept (Eurocontrol 2023), we designed a U-Space for Stockholm (2030–2040) featuring a centralised UAM Control Centre to ensure safety. This centre will be staffed by a human UAM Coordinator working alongside the digital assistant, DUC. Together, they will manage airspace and provide U-Space services. The U-space environment envisioned for Stockholm, the UAM Coordinator role, and the DUC are all futuristic concepts that do not yet exist, posing challenges in gathering needs and requirements for DUC explanations due to the absence of current users. The UAM Coordinator and DUC are envisioned to handle tasks such as traffic monitoring, flow management, ground-air coordination, and emergency response in the U-space. They must address the unique challenges of low-altitude urban airspace, coordination with ground activities, and managing diverse vehicles like electric vertical take-off and landing vehicles and drones. A more detailed description of this concept can be found in Carl Westin et al. (2025) paper (Carl Westin et al. 2025).

The UAM Coordinator will serve as a key safety function responsible for managing traffic within U-space airspace, ensuring the protection of both airspace users and the public. A central aspect of the UAM Coordinator's role will be collaborating with other human stakeholders to address problems and unexpected situations, including emergencies. Supported by the DUC, the UAM Coordinator will supervise U-space operations and maintain coordination with essential stakeholders. The DUC will handle routine operational tasks such as flight authorisation, traffic and conformance monitoring, information dissemination, and alerting the UAM Coordinator to critical events through visual interface cues.

In addition to these duties, the DUC should be capable of providing actionable recommendations to the UAM Coordinator for resolving emerging issues. These may include traffic rerouting, optimising traffic flows, resolving conformance deviations, and managing emergency scenarios. Emergency situations could

involve emergency landings, adverse weather, medical events onboard air taxis, ground disruptions affecting airspace (e.g. fires or smoke), or communication link losses. In such cases, the DUC will notify the UAM Coordinator, who will then intervene as needed. These interventions may require the UAM Coordinator to carry out tasks such as adjusting geo-fences, coordinating medical diversions with hospitals and emergency services, supporting search and rescue operations, or liaising with Air Traffic Management (ATM) for rerouting flights that involve airport coordination.

A key requirement for effective human-AI teaming is automation transparency and that the human operator can understand the AI-based systems' outputs (National Academies 2022; Endsley 2023). DUC must be capable of providing human-understandable explanations of its reasoning process, including how specific outputs are derived. This necessitates the DUC's ability to expose its underlying decision-making logic in a form that supports end-user comprehension and trust.

3. Theoretical background

This section examines the research areas of human-AI teaming and Explainable AI (XAI), exploring how the integration of storytelling and the Construal Level Theory (CLT) framework can support the design and structure of explanations through narratives.

3.1. Human AI teaming and explainability

Recent research highlights the importance of designing effective human-AI teams, especially in complex fields like military and aviation (National Academies 2022). These teams, comprising at least one human and one AI agent, can outperform either alone (O'Neill et al. 2022). Success depends on users' ability to understand, predict, and develop appropriate calibrated trust in AI, with key factors including transparency, explainability, and situational awareness (National Academies 2022). Trust can be defined as 'the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability' (Lee and See 2004). In this context trust does not mean unconditional reliance but rather an informed confidence that helps the user to critically assess when to accept or question AI recommendations. In safety critical domains, both over-trust or under-trust can be problematic (Tversky and Kahneman 1981; Endsley 2023). That is why an effective XAI should aim for a calibrated trust, the ability of the user to judge when to question or when to rely on the system. Moreover, how information is presented can influence trust calibration. As shown by Tversky

and Kahneman's (Tversky and Kahneman 1981), the way information is framed affects how people perceive and evaluate choices. Similarly, in human-automation interaction, the framing of AI explanations can shape users' perceptions of reliability and influence whether they place appropriate or excessive trust in the system.

Transparency, defined as AI's ability to enable understanding and prediction (Westin, Borst, and Hilburn 2016), varies in levels – from basic status updates to explanations of reasoning and uncertainty (Chen et al. 2014). Endsley (Endsley 2023) makes an important distinction between transparency and explainability: transparency supports real-time situation awareness by helping the user understand what the system is doing in the moment, while explanation serves a retrospective function, helping users build accurate mental models of the AI over time. XAI aims to enhance transparency by providing comprehensible AI decision explanations, fostering trust and informed decision-making. Approaches include interpretable models (e.g. decision trees) and post-hoc explanations using visualisations, examples, or natural language summaries (Guidotti et al. 2018; Xu et al. 2019; Shi et al. 2023).

To assist people in comprehending, trusting, and interacting with AI systems, several transparent and XAI approaches have been used for user-centered reasons in a variety of sectors (Lai et al. 2023). These methods include example-based explanations (Keane and Smyth 2020; Tschandl et al. 2020), counterfactual explanations (Wachter, Mittelstadt, and Floridi 2017), interactive visualisations (Krause, Perer, and Ng 2016), and model-agnostic approaches (Ribeiro, Singh, and Guestrin 2016). Although these methods are proven to be successful, they have their limitations in combining engagement, memorability, and retention of understanding that the storytelling approach could achieve (Martens et al. 2024). According to Miller (Miller 2019), human users frequently look for explanations that are both truthful and relevant to their circumstances. Explanations that follow a familiar, narrative format are easier for humans to grasp, which is why storytelling may be suitable for user-centered AI explanations (Martens et al. 2024). By placing information within a context, stories aid in situational awareness, which is crucial for users who want clarification in high-stakes situations like aviation (Fisher 1984; Hoffman et al. 2018).

3.2. Storytelling as a method for knowledge sharing and presenting explanations

One of the oldest forms of human communication is storytelling. Through the history of humanity, storytelling and narratives have been applied across numerous

disciplines. (Yilmaz and Cigerici 2018). Storytelling has long been recognised as an effective method for information transfer and for presenting explanations, at both individual and societal levels (Connell, Klein, and Meyer 2004; Koenitz 2023). Beyond psychology and ergonomics, where storytelling has often been studied as a tool for enhancing comprehension and retention, extensive traditions of research in the arts and social sciences have demonstrated its enduring relevance (Stache 2017). Storytelling has been used in communication design and marketing to engage audiences and shape interpretation (Dessart 2018); and in arts-based inquiry to support creative exploration and reflexivity (Butler-Kisber 2018). For speculative and design research, storytelling plays a central role; design fiction, for instance, uses narrative-driven 'diegetic prototypes' rooted in plausible real-world contexts to explore future possibilities and provoke reflection on present assumptions (Vistisen 2021). Contemporary examples of interactive digital storytelling appear in games, animations, films, and immersive experiences. Its power lies in its ability to organise information in narrative structures that align with natural cognitive processes, capturing attention and improving understanding (Bruner 1991). Research shows that storytelling helps users develop a strong emotional connection with the content and improves retention (Dahlstrom 2014). By presenting complex information in a narrative format, storytelling activates the brain's natural ability to process stories, making content more memorable and engaging (Segel and Heer 2010). Framing storytelling in this broader interdisciplinary context emphasises that this established, and cross-disciplinary practice and approach can be adapted to novel domains.

Storytelling techniques have shown success in explaining complex topics in a more engaging way, particularly in the context of data communication (Segel and Heer 2010). Researchers have explored how AI can support storytelling by authoring, developing, and generating interactive narratives, resulting in tools to assist content creation (Stefnissou and Thue 2018; Thorne 2020; Hernandez, Bulitko, and St. Hilaire 2021; Swaroopa 2024). While much research has focused on using AI to generate narratives, not nearly as much empirical work has explored storytelling approaches to improve users' understanding of AI itself. A case study exploring the use of narrative-based explanations for AI model outputs found that over 90% of surveyed participants including members of the general public and data scientists perceived the explanations as convincing, suggesting that storytelling can enhance understanding of AI predictions (Martens et al. 2024). While the study focused on non-expert and semi-expert users, the authors emphasised

the need for future research to examine how such explanations are received by domain experts.

As part of the ‘XAINES’ project investigating explanations of AI using narratives, Hartman et al. (Hartmann et al. 2022) emphasise the importance of tailoring narratives to end users’ needs (in their case people without machine learning knowledge), using techniques like vocabulary adaptation, interactive dialogs, and domain-specific examples. The paper gives an example of the application to healthcare professionals, particularly diagnostics like justifying model predictions from medical images such as X-rays. However, they do not disclose any details of the implementation or empirical evaluations of its usefulness.

McDermott & Folds (McDermott and Folds 2022) proposed that the most effective way to provide explanations is through a ‘narrated story approach’ that integrates both overarching goals and specific steps. They created a framework combining storytelling and CLT for structuring explanations across six levels, ranging from concrete to abstract. According to CLT, people interpret events at different levels of abstraction, from concrete (low-level) to abstract (high-level) construal (Trope and Liberman 2010), where construal refers to how individuals perceive and interpret information. CLT also applies to digital communication, explaining why individuals may feel more connected to a distant partner while messaging than to physically present friends (Norman, Tjomsland, and Huegel 2016). McDermott & Folds (McDermott and Folds 2022) applied their framework to a complex military command and control system, structuring explanations at different stages: during planning (ahead of operations), in real time (during operations), and post-operation (for review). Their example involved large teams, including a commander, multiple ships, and operators overseeing unmanned vehicles. While CLT guided the level of detail in explanations, the information was conveyed through storytelling, incorporating images, videos, and narration. They used storytelling elements, including actors, settings, and plots, to create a passive experience rather than requiring interaction with a map. Table 1 provides an overview of their CLT model, with a

description of each level, presentation duration and phase. While they argued this would reduce cognitive load, their framework and design were not evaluated with end users. The authors stress the importance of matching information presented in the ‘narrated story’ to the needs of the user:

Too much unwanted detail can prompt users to lose interest, and not enough detail can produce frustration, especially if the missing details are needed for task performance (McDermott and Folds 2022).

4. Method

This section describes both the participatory design process of the storytelling explainer and its subsequent human-in-the-loop simulation. It first introduces the U-space traffic scenario and the UAM Coordinator’s working position, as these constitute essential constraints and prerequisites for the design of the storytelling explainer. It then details the methodological framework and choices applied in the simulator experiment.

4.1. Scenario

Three Air Traffic Management and U-space domain experts from the Swedish Air Navigation Service Provider LFV helped define a novel U-space airspace and scenario used in the study, based on the CORUS-XUAM U-space concept definition (Eurocontrol 2023). The scenario took place in the Stockholm U-space airspace comprising an emergency event with an air taxi requiring rerouting to a hospital due to an onboard medical emergency: The following description of the scenario was used in the study:

‘An air taxi that is flying from Globen to Täby Centrum in Stockholm finds out that a passenger near Hammarbyslussen has passed out during a medical emergency (heart attack). The air taxi operator gets an alert, and with the help of the digital assistant, DUC. DUC proposes three hospitals for a reroute and UAM coordinator has one minute to decide.’

4.2. The Digital Assistant ‘DUC’ Concept and Implementation

The digital assistant, DUC concept was developed to represent an autonomous agent capable of engaging in a collaborative process with the human (the UAM Coordinator) to management U-space traffic, The DUC concept adhered to the EASA Level 2B classification of AI-based systems designed for human-AI teaming (EASA 2024). In the scenario, DUC tasks and

Table 1. CLT description.

CLT	Description of information	Duration	Phase
1	Executive summary, main claim	≤ 10s	During an event
2	Executive mission review, main reason	≤ 30s	During an event
3	Mission summary, justification	≤ 5 min	During an event
4	Mission brief, basis of justification	≤ 30 min	During an event
5	Mission plan, summary of data	≤ 60 min	After an event
6	Mission details, all data	> 60 min	After an event, on demand

actions were scripted beforehand on a timeline making each interaction from DUC appear at a specific time point. DUC would first present a dialog window describing a medical emergency that has occurred. When the medical emergency occurred, DUC proposed three alternative rerouting options for the UAM Coordinator to consider and decide on (Figure 1). Each alternative led to a specific hospital. Every alternative had a colour indicating the priority of that selection.

Several parameters including a) waiting list, b) vertiport availability, and c) cardiologist availability were considered when indicating the availability of each alternative (e.g. hospital), which in turn indicates the priority. The green colour was designed to indicate the highest priority, yellow with a lower priority, and the red dot indicates the lowest priority. The green dot next to 'Karolinska' indicated the preferred choice suggested by DUC. An initial concept was developed based on glyph design principles from Nylin (2023) (Nylin 2023). The glyph displays the speed and altitude of an Air Traffic Controller (ATCO). A decision time was then added to the glyph, as well as a symbol to define the frame of the situation (Figure 1).

4.3. UAM Coordinator Working position

The UAM Coordinator's working position was created using a portable simulator platform with two external screens (Figure 2). The primary screen (left) was an interactive U-space situation display interface. It served as the main screen for the UAM Coordinator to monitor U-space operations and interact with the DUC. Developed on the UTM City platform (Westberg, Palmerius, and Lundberg 2022) and aligned with the European CORUS-XUAM concept (Eurocontrol 2023) for the 2030–2040 timeframe, it provides a 2.5D satellite-

based bird's-eye view of Stockholm's U-space on a high-resolution display, supporting both detailed and city-wide situational awareness. The second screen (right) was dedicated to the storytelling explainer, presenting video-based explanations. Since the DUC was conceptual participants interaction with the assistant was facilitated through a Wizard of Oz approach, where an experimenter activated pre-scripted communication from DUC that appeared in dialogue windows on the U-space interface.

4.4. Design Process: creating the visual narrated storytelling concept

The explainer was developed to support first-time encounters with U-space traffic management by introducing the novel role of the UAM Coordinator and their collaboration with the conceptual digital assistant, DUC. Given that the UAM Coordinator role is novel, proposed as a needed human role in U-space traffic management, the explainer was framed to resemble an initial training context, where operators are familiarised with new systems and responsibilities, in particular the DUC assistant.

The storytelling concept was developed to support UAM Coordinators by providing information on system functionalities and decision-making processes. The following design aspects were considered: CLT level (detail and duration), modality (text, visual and audio), and temporal aspects (when to present the storytelling explanation). The explainer was designed to be shown on a separate screen from the operational interface, a choice intended to benefit the structure of information in the work position. This spatial separation enables users to focus on the narrative explanation without cluttering

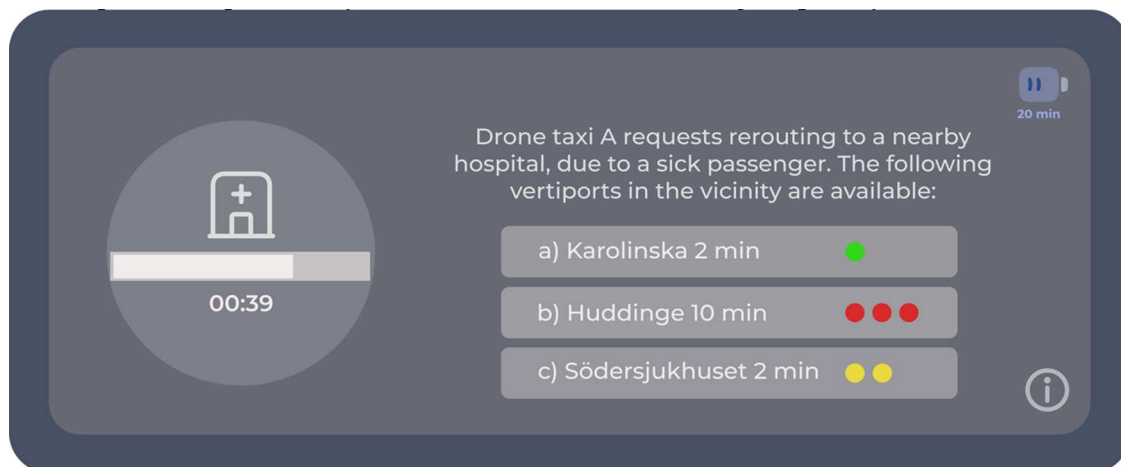


Figure 1. DUC's solution glyph with the hospital rerouting recommendations in the medical emergency.

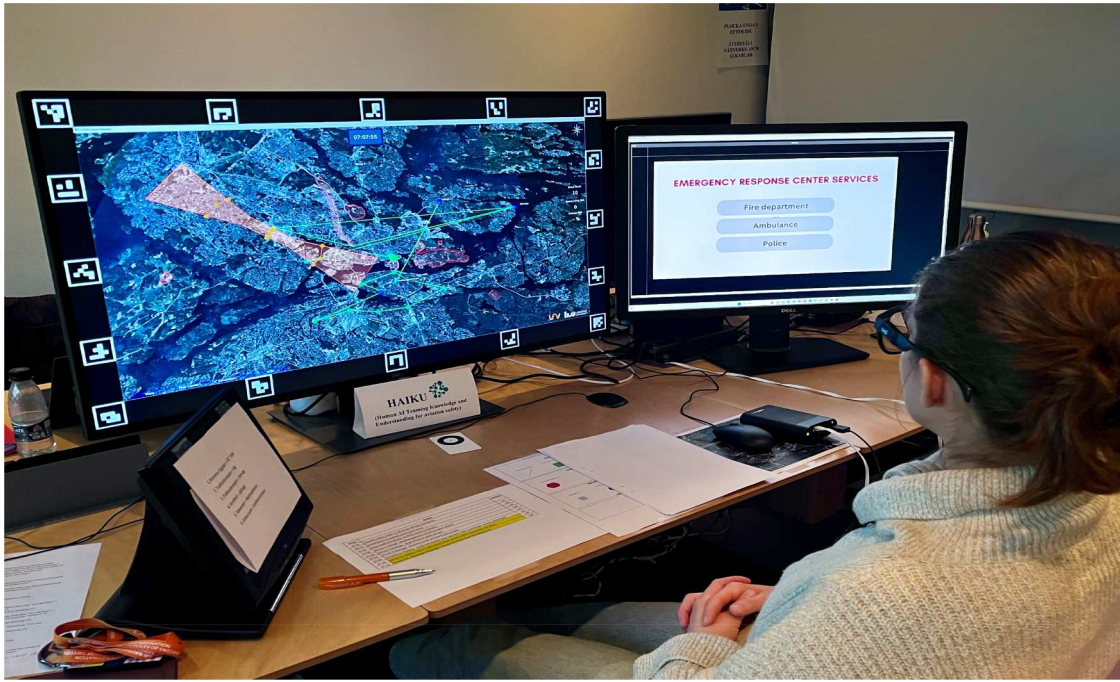


Figure 2. UAM Coordinator working position with the U-space interface (left) and storytelling explainer (right). The interface visualised key U-space airspace elements including aircraft (green dots), U-plans (green flight trajectories), airspace constraints (red and orange geo-fence zones), and vertiports (cyan squares). These elements had resemblance to conventional airspace elements found on contemporary ATC radar systems. Users could navigate the environment through zooming and panning. Through dialogue windows and highlighted overlay labels, DUC provided status updates and alerts on events such as airspace changes, deviations, and emergencies.

the main task interface, thus maintaining clarity between system status and system reasoning.

Given the conceptual nature of the UAM Coordinator role, identifying user requirements posed challenges as there are no existing end users. Air Traffic Controllers (ATCOs) were determined to be the closest existing role to the UAM Coordinator, given that they manage traffic in conventional airspace. The design process followed an iterative structure incorporating workshops, interviews, and design iterations with three ATCOs and/or U-space domain experts to gradually refine the concept. The process began with an initial workshop that introduced the scenario, DUC's tasks, and the recommendations for explanations that were subsequently identified. This was followed by a series of four workshops and two interviews, where insights from U-space experts and ATCOs helped shape the design. Each iteration involved presenting and refining storyboards, prototypes, and high-fidelity designs, incorporating user feedback at every stage. The number of participants varied per session, with four ATCOs engaged throughout the process. Through this step-by-step approach, knowledge was progressively built, ensuring that each iteration addressed real operational needs. The final design emerged from this continuous refinement,

aligning with the practical requirements of ATCOs managing high-density urban airspace.

The visual narrated storytelling explainer, which was approximately three minutes long, explained the decision-making processes of DUC, including how it derived the three hospital alternatives. The explainer appeared several minutes before the first pop-up window with decision alternatives (glyph) to ensure that explanations were received before participants made a decision. Given the novel UAM Coordinator role, additional introductions to the context (U-space and UAM), the human role (UAM Coordinator) and the digital assistant (DUC) were created using a storytelling approach to be presented as introduction videos to the simulation exercise.

4.5. Design dimensions of the storytelling concept

The design focused on creating a storytelling explanation that was clear and suitable for end users by selecting appropriate levels of detail and timing, guided by the CLT framework. We used McDermott & Folds' storytelling format of a narrated story combining visual elements (video, animations, text, and graphics) with audible narration and captions to convey a cohesive story

(McDermott and Folds 2022). Video was chosen as the medium because it effectively combines visual and narration storytelling to enhance comprehension, engagement, and retention. The relationship between video and narration is essential – while visuals provide context and structure, narration complements visuals by explaining the decision-making process, ensuring a seamless cognitive flow. Subtitles were included to accommodate diverse user needs, ensuring accessibility in noisy environments or for users who prefer reading over listening.

Levels 1 and 2 followed McDermott & Folds' (McDermott and Folds 2022) duration recommendations (10–30 s), while Levels 3 and 4 were shortened to 1–2 min (instead of 5–30 min) to better suit the context and time-critical nature of the medical emergency. Levels 5 and 6 were excluded as their durations (+60 min) were found unsuitable for implementing explanations to support decisions in real-time situations (Table 2). Slide examples from the storytelling explainer for different CLTs can be seen in Figure 3.

In the working position setup, the storytelling explainer is shown on a separate screen next to the U-space situation display. The explainer is triggered by the system (i.e. DUC) when presenting recommendations on where to reroute the air taxi. The recommendations are presented in a glyph (i.e. pop-up window) on the U-space situation display. The glyph (Figure 1) also conveys key situational factors. The glyph content is the outcome of the DUC decision-making process that is shown and explained in the storytelling explainer, reinforcing the video-narration combination as an optimal way to communicate AI-driven decisions effectively.

4.6. Simulator experiment

The study used a mixed-methods approach. The quantitative part comprised a questionnaire with Likert-scale statements. The qualitative part collected data through observations and interviews.

Table 2. CLTs used with description.

CLT	Phase	Description	Time
CLT1	Executive Summary	3 hospital options to reroute air taxi with sick passenger.	30 s
CLT2	Mission Overview	Overview of DUC's decision-making in a medical emergency, key stakeholders involved, and its communication channels for receiving information.	
CLT3	Mission Summary	Aspects analyzed to generate three hospital reroute options.	2:15 min
CLT4	Mission Brief	Detailed breakdown of aspects analysis in relation to different hospitals, considering time to destination, waiting lists, vertiport and cardiological department availability.	

4.6.1. Participants

Eight licensed Air Traffic Controllers (ATCOs) from Malmö Air Traffic Control Centre participated who were assumed to be the closest profession to the future role of the UAM Coordinator. Based on this, the ATCOs were the first-time users of the system (DUC). Their ages ranged from 24 to 57 years ($M = 42.6$), with experience varying from 0.4–34 years ($M = 16.9$). The non-probability sampling method was based on a convenience sampling with voluntary participation. While all participants had a basic understanding of UAM, they lacked prior knowledge of the DUC and the UAM Coordinator role.

4.6.2. Procedure

Each participant completed a three-hour simulation, beginning with a consent form and demographics questionnaire (age, experience, performance ratings). While the simulation contained several scenarios, the focus of this paper is only the medical emergency scenario. Participants played the same medical emergency scenario twice, once with a storytelling explainer for the DUC's recommendations and once without. Each run lasted for around 30 min; the order varied to mitigate learning effects. The scenario gradually increased in traffic complexity by incrementally adding U-space elements like no-fly zones, vertiports and drone traffic. Participants were briefed on DUC's autonomous role and tasks, such as activating U-plan, traffic conformance monitoring, and offering rerouting suggestions. Since the DUC was conceptual participants interaction with the assistant was facilitated through a Wizard of Oz approach, where an experimenter activated pre-scripted communication from DUC that appeared in dialogue windows on the U-space interface.

Eleven minutes after scenario start, the DUC alerted the UAM Coordinator to a medical emergency via voice prompts. Thirty seconds later, DUC presented a pop-up window with recommendations on three hospitals where the air taxi could be rerouted, colour-coded by priority (green = best, red = worst). The DUC recommended Karolinska (green), factoring in vertiport availability, waiting lists, and travel time. Before deciding, participants were instructed to watch the storytelling explainer. They then selected a rerouting option by clicking on a glyph. After the scenario had finished, participants answered a questionnaire with Likert scale statements assessing understanding, trust, and perception of DUC after the storytelling explainer and user experience questions (User Experience Questionnaire (UEQ) 2024). Questionnaire items are shown in Figure 4.

Quantitative data were analyzed in Excel, focusing on medians (for central tendency) and interquartile range

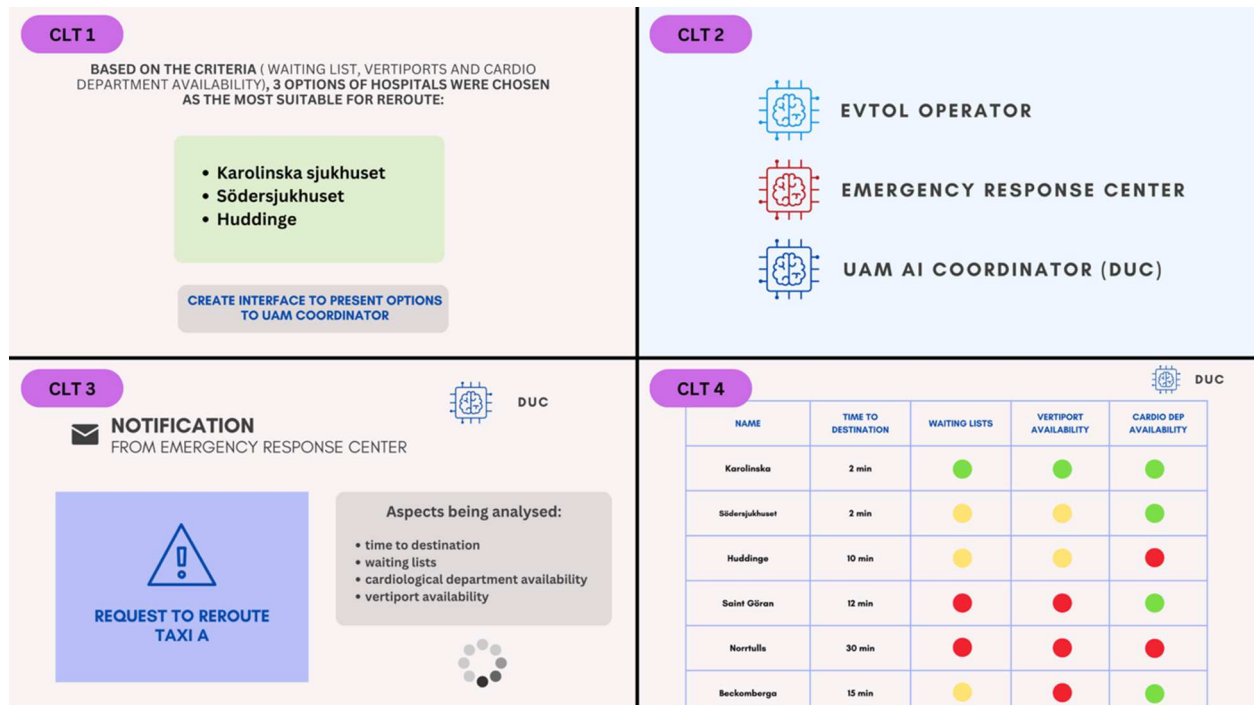


Figure 3. CLT 1–4 visual examples from the storytelling explainer.

(IQR, for dispersion). The debriefing covered five areas: general storytelling experience, understanding of explanations, trust in DUC, cognitive load, and improvement suggestions (Appendix A). Interview and observation data were transcribed and analyzed thematically. We used a summative evaluation with retrospective questioning, as opposed to a formative approach, such as in-situ prompts, that could have disrupted task performance, to preserve situational awareness and workflows during runs. The thematic analysis followed Braun and Clarke's (Braun and Clarke 2006) six-phase method. Two researchers conducted the first four phases independently. The process began with the transcription of eight recordings, which were automatically transcribed using Whisper software. In the second phase, the transcriptions were coded, resulting in 88 initial codes. The third phase involved searching for themes, which produced nine preliminary themes. In the fourth phase the themes were reviewed and refined, resulting in a final set of six themes. During the fifth phase results were compared, discussed, and theme labels were defined. Finally, the report was produced. The themes were generated inductively from the interview data, without applying predefined theoretical categories. This ensured that the major categories emerged directly from participants' accounts rather than being imposed from existing frameworks. The final set of themes was therefore entirely data-driven and grounded in participants' perspectives.

5. Results from the simulator experiment

The study used a mixed-method approach in the form of both quantitative and qualitative data collection, to get insights into the visual narrated storytelling explainer. Quantitative results from Likert-scale questionnaires, complemented by qualitative data from interviews.

5.1. Questionnaire

The questionnaire results in Figure 4 presents a box-and-whisker plot illustrating participants' ratings of the storytelling explainer across various dimensions. The pink bubbles represent individual participant responses, providing a visual indication of the spread and distribution beyond the median, interquartile range, and overall range.

Participants rated the storytelling explainer as friendly ($Mdn = 5$), clear ($Mdn = 4.5$), interesting ($Mdn = 4$), easy ($Mdn = 5$), motivating ($Mdn = 4$), and meeting expectations ($Mdn = 4.5$). However, terms like 'complicated/easy', 'boring/exciting' and 'slow/fast' have more spread-out ratings, indicating a mix of opinions. Participants were found to agree with the statement that the story contributed to the interaction with DUC ($Mdn = 4$). Responses regarding collaboration with digital assistant 'DUC' and how explaining stories contributed varied widely, with some participants rating them highly while others rated them very

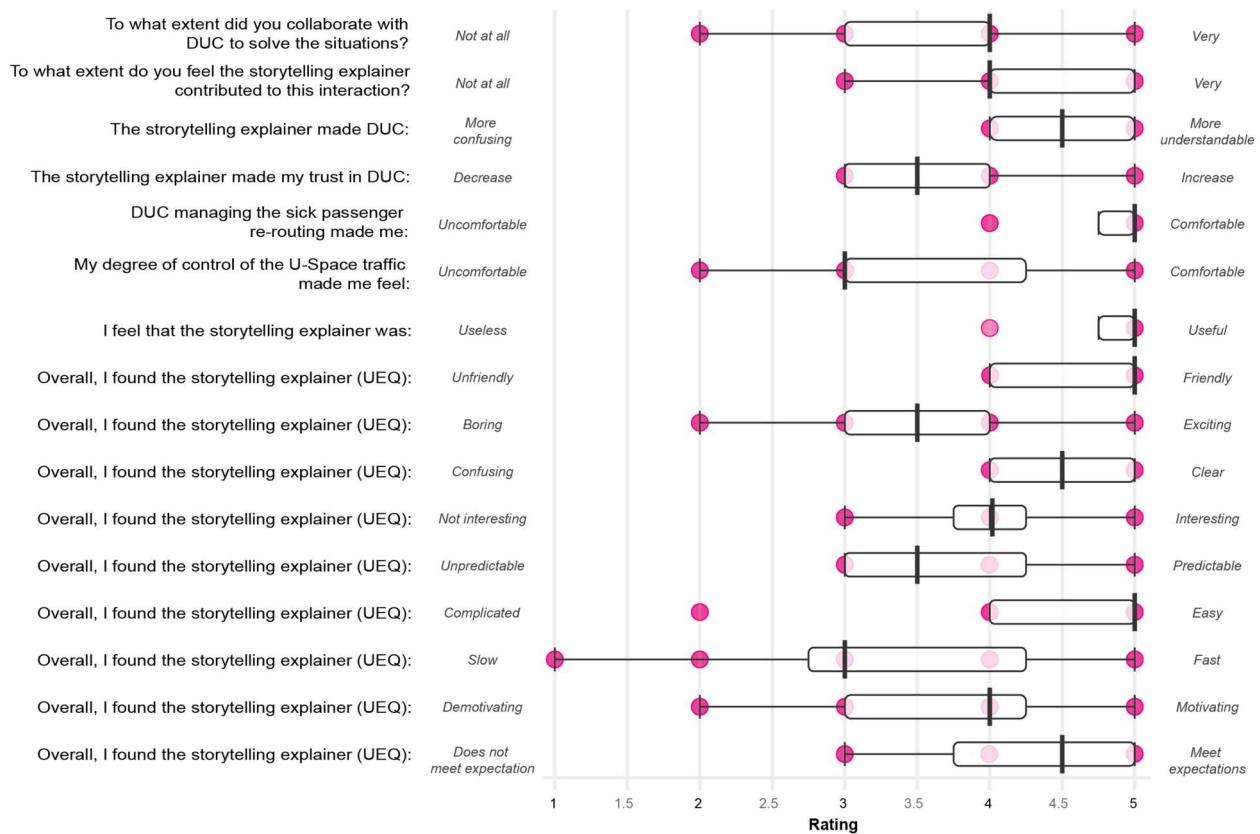


Figure 4. Storytelling questionnaire. Items had scales from 1 to 5, where 1 is negative, 5 is positive and 3 is neutral.¹

low. Similarly, participants were found to agree with the statement that the storytelling explanation made DUC more understandable ($Mdn = 4.5$). The storytelling explainer was rated to marginally improve participants' trust in DUC ($Mdn = 3.5$). DUC's management of the sick passenger re-route was rated as a comfortable action ($Mdn = 5$). The storytelling explainer was rated as useful ($Mdn = 5$).

5.2. Debriefing results

Thematic analysis of the interviews, conducted following Braun and Clarke's framework (Braun and Clarke 2006), resulted in a rich set of themes reflecting participants' experiences with and perceptions of the storytelling explainer. The findings are organised into the following major categories: (1) Perceived Format Benefits, (2) Trust and System Understanding, (3) Challenges of the Storytelling Format, (4) Interactivity and Control, and (5) Reflections on Modality and Presentation. The distribution of quotation codes across five categories is shown in Table 3.

5.2.1. Perceived format benefits

Participants described the storytelling explainer as easy to understand, helpful, and cognitively accessible. Two

participants (P5 and P8) explicitly emphasised that the narrative structure aided their understanding, particularly as first-time users of the system. In addition, six other quotations across the dataset referred to the storytelling format as 'easy to understand,' 'clear,' or 'helpful for clarifying important aspects,' suggesting that most participants perceived the narrative structure as beneficial for supporting understanding. Storytelling was seen as a way to simplify complex decisions and communicate system reasoning in a relatable format. One participant reflected positively: 'It was very basic, which I like because then it's more understandable.' (P8) Explanations were clear and detailed enough and helpful to have at same time as event happens. The integration of a realistic medical scenario helped situate the system's decision-making in a meaningful context, aiding decision support. Three participants felt the storytelling format

Table 3. Distribution of quotation codes across major categories.

Major categories	Total coded quotation
Perceived format benefits	28
Trust and system understanding	16
Challenges of the storytelling format	18
Interactivity and control	16
Reflections on modality and presentations	15
Total	93

enhanced their ability to select the most appropriate hospital reroute option. One of them stated: 'I think it was good information. I felt more interested and more understanding about it. And I felt like this is good technology. Absolutely helpful.' (P6) Others emphasised that the medical narrative was particularly beneficial for solving the rerouting problem and clarifying which hospital was most suitable. The format also seemed to be suitable for early users of the system and reduces the perceived need for technical manuals or lengthy documentation. As one user remarked: 'As a beginner, it's very good to have the explanation of everything.' (P5) Participants further noted that the explainer's clarity helped them efficiently extract relevant information: 'It was informative, and it was easy to pick the important stuff and what you needed to sort out.' (P3)

5.2.2. Trust and system understanding

Four participants reported that the explainer improved their perception of the digital assistant, DUC as reliable. The storytelling explanation of how DUC generated its recommendations contributed to a sense of transparency and logic in the system's reasoning. For example, one participant said: 'For the information I got from the stories, I thought the DUC was reliable.' (P3). While another emphasised: 'I think it made it more trustworthy. Like when you get a bit more knowledge about how it works, you trust it a bit more.' (P8) Others, however, expressed that full trust remained conditional on a deeper understanding of the system's limitations and decision boundaries. 'I trusted it more but not completely. I wanted to see what it doesn't take into consideration.' (P4) Importantly, one participant noted that DUC's presentation of alternatives supported their decision-making but also implicitly required trust due to the lack of alternative sources.

5.2.3. Challenges of the storytelling format

Despite general support for the storytelling approach, participants raised concerns about information overload and unclear content boundaries. Three participants noted that too much information, particularly medical details, were unnecessary or difficult to process under time pressure. 'And now it was all about the heart rate ... And I don't want that information. I just want to know if it's a critical amount.' (P5) Five participants reported that they were confused and did not clearly understand how DUC functioned, despite the explanation, indicating a need for further clarity in how system logic is conveyed. There was also concern about the suitability of storytelling in highly time-sensitive or situationally intense scenarios, with some suggesting it

was impractical to follow detailed narratives under cognitive load. According to three participants, it promoted passive human involvement, reducing active engagement. Four participants also discussed the difficulties relating information in storytelling explainer to situation on map and that explainer did not support geographical orientation.

5.2.4. Interactivity and control

One of the most prominent themes was the lack of interactivity in the storytelling explainer. Participants described the experience as overly passive and expressed a strong desire for more control over how and when explanations were accessed. 'It wasn't much of an interaction actually. It was more like a story told and information, so no, like, interactions. I could choose the options, but it didn't feel like an interaction actually.' (P5) Multiple suggestions were made for improving user agency. These included allowing users to request explanations on demand, add a replay function, and offer customisation of the explanation level of detail. For example: 'It would be maybe even better to have it in sound, like a communication, where you can ask questions.' (P7) 'There was this information button on the pop-up messages and that you can ask like stories to appear on request.' (P1) Such features would allow users to tailor the storytelling to their current workload and information needs particularly useful in operational settings with fluctuating cognitive demands.

5.2.5. Reflections on modality and presentation

Participants expressed differing preferences regarding how explanations were presented. While video and narration were generally favoured for their clarity and engagement, three users noted that narration was impractical during high workload situations, especially if they missed a part of the story and could not easily rewind. Two participants suggested formats that would allow rapid scanning, such as text, or suggested combining visual storytelling with readable summaries for flexibility. Although another two participants thought the chosen modality was a suitable choice: 'Seeing this is better than reading something ... when you read it, it's easy to miss out on stuff. But when someone tells you, I think it's easier to pick up the information.' (P8) Others emphasised the importance of mapping the explainer's content more closely with what was shown in the simulator interface, to reduce split attention and improve coherence between storytelling and the traffic display.

6. Discussion

This study sets out to explore (1) how a storytelling concept using visuals and narration can be designed to convey system explanations in a decision-making task, and (2) how such explanations affect users' understanding and attitudes toward explanations. In the following subsections, we begin by outlining key insights gained from the design process of the storytelling explainer. We then examine the findings from the end user evaluations, discussing their implications for the design of explanation presentation and for effective human-AI teaming. In the subsequent section, the key insights and lessons learned are consolidated into design recommendations for leveraging storytelling to effectively present explanations. Finally, we discuss the limitations of our study. This work builds on and extends prior storytelling research by empirically examining its application for explaining AI outputs in a safety-critical domain – a setting where human understanding is essential to operational trust and performance. Prior work, such as McDermott and Folds (McDermott and Folds 2022), Martins et al. (Martens et al. 2024) and Hartman et al. (Hartmann et al. 2022), has proposed storytelling frameworks for XAI and highlighted the potential of narrative explanations. However, these approaches (Hartmann et al. 2022; McDermott and Folds 2022) have largely remained conceptual or were demonstrated in non-operational settings, with limited or no empirical validation. Martens et al. in particular recommended that future research investigate how storytelling explanations are perceived by domain experts, a gap this study directly addresses by involving licensed air traffic controllers acting as UAM Coordinators. Thus, our findings contribute empirical evidence to the evolving research on storytelling-based XAI in high-stakes environments.

6.1. Design of storytelling explainer level of detail in explanations

This subsection elaborates on the first research objective: *how a storytelling concept using visuals and narration can be designed to convey system explanations in a decision-making task.*

6.1.1. Using storytelling to convey explanations

The tone of the visual materials was also an important factor. The explainer avoided cartoon/childish-like aesthetics, instead employing an infographic style appropriate to the professional context of airspace management. In high-stakes environments such as aviation, visual seriousness can reinforce the critical nature

of the task (McCloud 1993). The storytelling explainer used an adaptive (complementary) approach for combining visual and narrative elements. The strength of doing so is to afford each medium the space to leverage its unique strengths to improve communication: visuals are powerful for conveying physical details and evoking emotional responses, while text offers clarity, precision, and the ability to express abstract concepts. However, there are different combinations that can be used, and researchers should consider which combination is most suitable for their needs. For instance, it might be desirable to use a combination where both the narrative information and visual information convey the same story (a duo specific combination), ensuring that it is sufficient to attend to one source for understanding the explanation.

6.1.2. Application and adaptation of the construal level theory layered approach

The CLT layered approach allowed users to gradually build an understanding without being overwhelmed, and it mirrored the different cognitive demands present at various stages of a decision-making process. While CLT has been previously proposed as a theoretical framework for explanation design (McDermott and Folds 2022), this study applies and evaluates it empirically in a novel context. Unlike McDermott and Folds' study (McDermott and Folds 2022), which focused on planned missions and retrospective analysis in a multi-actor command-and-control setting, our study focused on real-time decision support in a training context. While their framework addressed explanation needs for multiple actors with varying requirements, we focused on a single human operator, the UAM Coordinator, collaborating with the AI assistant DUC to make immediate decisions. McDermott and Fold describe CLT 5 as having an expected duration of nearly an hour and CLT 6 as providing highly detailed, on-demand information, with both levels appropriate for post-operation review. For this reason, we excluded CLT 5 and 6 as they were deemed unsuitable for time-sensitive traffic monitoring as they do not align with the fast-paced emergency context of U-space operations. Instead, we adapted CLT to classify explanations by level of detail while ensuring their relevance to real-time decision-making.

6.2. User experience with the storytelling explainer

This section addresses the second research objective: *to evaluate how explanations presented through visual narrated storytelling affect users' understanding and*

attitudes toward a conceptual digital assistant system in a real-time decision-making context.

Findings from the end user evaluations indicate that participants generally perceived the storytelling explainer as user-friendly, clear, interesting, easy to use, and motivating, aligning with their expectations. However, some participants found the storytelling pace to be slow with a few considering it boring and predictable. Findings suggest that the storytelling explainer supported participants' understanding of DUC's decision-making, particularly regarding the hospital reroute logic. Participants described the explanations as clear, structured, and easy to follow, highlighting how the narrative format helped them grasp DUC's functions and rationale. The use of visuals and narration, combined with CLT-based structuring, facilitated comprehension by presenting information at appropriate levels of abstraction.

Effects on participants' attitudes such as trust in DUC were varied. While some participants reported increased trust following the explainer, questionnaire results were neutral overall. This may reflect the short exposure time, the conceptual nature of DUC, and the unfamiliar context. Notably, participants who encountered the storytelling explainer in the second simulation run expressed stronger positive reactions, suggesting that storytelling may be especially valuable as a contrastive explanation. Overall, while the explainer showed potential to support trust and system transparency, broader research is needed to understand how different explanation formats shape user attitudes across tasks, domains, and experience levels. The goal of explainer is not merely to increase trust but to help users make an appropriate decision. Users should be able to recognise when it is possible to rely on the system and when it is better to question it. It is also important to recognise where trust exceeds the system's capabilities. AI and human are two actors who can access information from different sources and work together to make the best decision, although the final decision belongs to humans. When users rely on the system despite such gaps, they risk over-trusting it which may lead to misuse and wrong decisions. How explanations are presented can influence this balance. The Framing Effect (Tversky and Kahneman 1981; Lee and See 2004) shows that the way information is framed affects judgment, and similarity could be observed in human automation interaction where presentation style shapes the reliability. Storytelling explainers must therefore be designed carefully to support appropriate trust, as one participant noted, full trust required understanding 'what it doesn't take into consideration,' underscoring the need for explanations that communicate not only what the system does well, but also its uncertainties and boundaries.

Debriefing results show that participants appreciated the level of detail on DUC's decision-making process conveyed in the storytelling explainer, although comments questioned the relevance of medical details. The study did not evaluate participants' perceptions of individual CLT levels or their relative value. Future research should explore separating these levels to determine their effectiveness for understanding system outputs and influence decision-making.

Findings furthermore suggest that storytelling-based explanations can be made effective by placing them in operational situations. Participants reported that visuals showing the emergency scenario supported their understanding of DUC's recommendations by enabling them to evaluate whether the system's interpretation of the situation matched their own. Narration, presented in natural and domain-relevant language, further facilitated comprehension.

6.2.1. Adaptive and on-demand storytelling

We underscore the need for AI systems to be adaptive and responsive to user-initiated requests for explanations. In operational settings, explanations should be available only when the user actively seeks clarification about the AI system's output (National Academies 2022). Findings from the study show that fixed-timing delivery, while useful for experimental control may disrupt workflows and cognitive flow in real-world operations. Participants expressed a clear preference for being able to access explanations when needed, rather than having them appear automatically. On-demand access, for example via a button on the interface, would allow operators to align explanations with their own task timing and workload. These observations point to broader design implications, including the importance of synchronising the content of the storytelling explainer with what is shown on the traffic situation display. In our simulation, the explainer was triggered at a predetermined moment to ensure consistency and simplify implementation. Supporting this type of user-driven interaction will be critical for integrating explainable AI into dynamic, time-sensitive environments.

6.2.2. Suitability for training vs. real-time operations

Future research should further refine the storytelling concept to improve its effectiveness in dynamic, high-stakes decision-making environments. We find that the use of storytelling as a format for explanations is valuable during learning and initial system exploration, particularly in simulator training environments where operators can familiarise themselves with the system without time constraints. The usefulness and

effectiveness of the storytelling format during real-time operations remain unclear. In time-critical situations, a visually narrated storytelling format that demands continuous attention may be impractical. While results indicate that storytelling could support end user's understanding of AI decision-making in response to live events, current design solutions for integrating it into time-sensitive contexts are lacking.

Future research should include comparative studies evaluating the storytelling format against alternative explanation formats, such as text messages and graphical icons or symbols. Additionally, further work should explore adaptive storytelling formats, such as multiple CLT-based versions of the same explanation, allowing users to choose between brief summaries for urgent situations or more detailed narratives when time permits. However, storytelling can be time-consuming, requiring sustained attention for several minutes, which is impractical in many high-pressure environments. Although storytelling could be integrated before, during, or after operations, defining its most suitable application remains an open question. Our findings suggest that background explanations not tied to immediate decisions may be more effective when delivered outside of critical moments, whereas in high-pressure situations, only brief and lightweight explanations are likely to be useful. Given the UAM Coordinator's limited time and cognitive capacity, any explanation format (storytelling or other) must balance informativeness with efficiency. Some participants suggested increasing interactivity in the storytelling explainer – for example, allowing users to request additional information, or to rewind and replay explanations if they missed something or wanted to learn more. This aligns with Endsley's (Endsley 2023) distinction between situation awareness and mental model building. Short, real-time explanations may support situation awareness in the moment, while richer storytelling explanations are more likely to contribute to longer-term mental model development by helping users understand how the AI system works and why it behaves as it does.

6.3. Design recommendations for using storytelling to convey explanation

Based on the design process and user evaluations, several key design recommendations have been identified to support the development of storytelling-based explainers for complex, high-stakes domains:

- Match visual tone to professional and operational context: Avoid cartoonish or overly playful aesthetics

to maintain credibility and trust (e.g. use of minimalist iconography and muted colours; structured typography; document style layout for medical details).

- Reinforce the critical nature of the task: Ensure visuals highlight the importance and urgency of the process (e.g. alert signals through colour and symbols; progressive escalating visuals; embedding direct medical data into the visuals)
- Consider how to combine visuals and narration in a way that thoughtfully leverages their individual strengths. For instance, a 'duo-specific' combination can be appropriate for work environments where operators are exposed to multiple, potentially competing visual and auditory inputs. In a duo-specific setup, both channels independently convey the core message. This enables users to understand the explanation through either listening or viewing alone.
- Avoid overly detailed or lengthy explanations and keep content concise to maintain user engagement and focus. Consider structuring explanations in layers: Present information progressively to help users build understanding gradually and prevent cognitive overload (e.g. Instead of overwhelming the viewer with details upfront, information is introduced in layers (CLTs); interface style summaries; consistent use of 'call to action' boxes/buttons to get to another part of the narrative).
- Think about how to deliver explanations without interrupting the primary interface. Video-narrated storytelling explanations are suitable to present on a separate screen or in dedicated window to prevent disruption of operational workflows.

Simulator experiment results further refined the design recommendations for interacting with the storytelling explainer and highlighted user perceptions of the system. While participants found the format generally helpful, they expressed a need for greater control and flexibility in how and when explanations are accessed.

- Support flexible, user-controlled explanation access: Provide options for replay, skimming, or switching to text summaries to improve efficiency and accommodate differing user needs.
- Treat personalisation as system capability: Allow the system to adapt narrative style and abstraction level based on operator expertise, task urgency, and cognitive state.
- Align explanation depth with CLT: Offer detailed narratives for onboarding and training, but concise summaries during high workload periods.

- Integrate narratives closely with spatial and visual context: Synchronise explanations with map and traffic display elements to improve comprehension and mapping between them.
- Offer user-controlled timing: Allow users to initiate explanations themselves rather than relying on automatic triggers, to better match individual workload and preferences.
- Provide adjustable detail levels: Enable users to select between brief summaries and more comprehensive narratives to reduce cognitive load and enhance clarity.

6.4. Limitations

While this study provides valuable insights into the use of storytelling for Explainable AI (XAI) in safety-critical human-AI collaboration, several limitations must be acknowledged. These relate to the sample size and participant profile, ecological validity of the simulation, the nature of the explanation content, and the lack of comparative evaluation. Each of these factors is discussed below.

6.4.1. Sample size and participant profile

The small sample size limits the generalizability of the findings. Only eight licensed Air Traffic Controllers (ATCOs) participated, all recruited under operational constraints that allowed only surplus staff to take part. Even then, participants could be recalled to duty, reflecting the broader challenge of conducting research in a domain where operational priorities outweigh availability for experimentation. As of 2024, Sweden had only 644 licensed ATCOs (Swedish Transportation Agency, 2024), making recruitment inherently limited. Furthermore, while the ATCO role shares functional similarities with the conceptual UAM Coordinator role such as airspace management and stakeholder coordination, key differences remain. ATCOs typically manage aircraft individually and proactively, whereas UAM Coordinators are expected to oversee system-level traffic with the support of automation. Future studies would benefit from broader and more diverse participant samples, including UAM trainees or professionals from adjacent domains, and could employ longitudinal designs to assess learning over time.

6.4.2. Ecological validity

A second limitation concerns ecological validity. The AI assistant 'DUC' used in this study was not an operational system, but a conceptual prototype implemented using a Wizard of Oz approach. As such, DUC's outputs were scripted and controlled by a human operator rather

than generated by a live AI model. While this limits realism, the Wizard of Oz method remains a powerful tool for early-stage design exploration. As noted by Dahlbäck et al., Wizard of Oz allows researchers to simulate system behaviour and test interaction concepts before technical implementation (Dahlbäck, Jönsson, and Ahrenberg 1993). In this study, it enabled the controlled evaluation of explanation strategies in realistic scenarios without relying on mature AI infrastructure.

6.4.3. Explanation content: expert-crafted vs. explainable AI-generated

The explanation content was designed by subject matter experts rather than derived from an actual XAI model. Although this ensured that the explanations were domain-relevant and accurate, it limits the representativeness of the system's reasoning as users would encounter it in real-world applications. Future research should incorporate XAI-generated outputs to assess how storytelling can be effectively layered onto live algorithmic decisions, especially in high-stakes or time-critical situations where fidelity and traceability of explanations are essential.

6.4.4. Lack of comparative evaluation

Finally, the study evaluated only one explanation format, visual narrated storytelling, without comparison to alternative methods such as textual explanations, static visualisations, or interactive dialogues. As such, it is unclear whether storytelling is more effective than other formats in supporting user understanding, trust, or decision quality. Comparative studies are needed to evaluate the relative strengths and limitations of different presentation modes, and to explore how storytelling might be integrated with interactive features that allow users to control the depth, pace, and timing of explanations.

7. Conclusions and future research

This study explored the design and evaluation of a storytelling explainer within the context of managing traffic in urban airspace, where a human operator collaborates with a digital assistant to manage an emergency. The challenge addressed through storytelling was how to facilitate the human understanding of the assistant's decision-making process. Findings indicate that some participants found the storytelling concept useful, as it supported their understanding and provided information at an appropriate level of detail. However, the temporal dimension must be handled more flexibly in learning-oriented explainers for mission-critical situations. In summary:

- The use of storytelling as a format for explanations is valuable during learning and initial system exploration, particularly in simulator training environments where operators can familiarise themselves with the system without time constraints.
- The storytelling concept was well-suited for presenting explanation about the digital assistant system's decision-making process.
- A key challenge is the temporal dimension of explanations, finding the right balance between the limited time available in critical situations and the mental effort required to understand the explanation. This balance may also depend on whether the user needs the information to make a decision at that moment.
- Interactivity could address the temporal challenges in storytelling explainers by allowing users to request additional information as needed.

As AI evolves to function as a teammate, the challenge is to ensure it remains understandable and trustworthy to human operators. While AI does not operate or 'think' like a human, explanations must bridge this cognitive gap. Decisions about how explanations are presented should consider how humans interact, reason, and develop an understanding of their surroundings. This study argues that approaches such as storytelling can make AI-driven decisions and behaviours more transparent and understandable.

Looking ahead, future research could explore how different storytelling techniques support user needs in various decision-making situations. This includes studying when users need real-time support to maintain situation awareness, versus when they need deeper explanations to build a mental model of the AI system. These needs often depend on how much time is available for the decision. To do this effectively, future work should move beyond general design heuristics like CLT and instead focus on modelling actual decision scenarios both with and without explainers to better understand their real impact on user understanding and performance.

Note

1. Some questions/statements were modified or shortened to improve clarity in this paper.

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Author contributions

CRedit: **Jekaterina Basjuka**:ConceptualizationData curationFormal analysisInvestigationMethodologySoftwareSupervisionValidationVisualizationWriting – original draftWriting – review & editing; **Carl Westin**:Project administrationSupervisionWriting – original draftWriting – review & editing; **Gustaf Söderholm**:SoftwareValidationWriting – review & editing; **Karl Johan Klang**:ValidationWriting – review & editing; **Jonas Lundberg**:Funding acquisitionProject administrationSupervisionWriting – review & editing

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Appendix A

Interview questions:

- Could you explain how does the DUC work based on the explaining story?
- After viewing the storytelling explainer, what are your initial thoughts or impressions about the storytelling format for receiving explanations?
- How would you describe your interaction with DUC?
- Were there specific aspects of the storytelling explainer that you found particularly helpful for understanding the DUCs decision-making process?
- Were there specific aspects of the explaining story that you found particularly confusing for understanding the DUCs decision-making process?
- How has the storytelling explainer affected your perception of how reliable/trustworthy the DUC is?
- Could you give an example of a part in the explaining story that was ‘too detailed’ / ‘too much information’ / ‘overload with information’?
- Can you give an example of a part in the explaining story that was ‘too abstract’ / ‘too little information’ / ‘had insufficient detail’ / ‘difficult to grasp’ / ‘too many details’?
- Can you give an example of a part that triggered you – something that you would like to know more about?
- Can you provide suggestions on how to refine how DUC communicates with you as the UAM Coordinator?
- How can the explaining story be improved to better meet the needs of the UAM coordinators in understanding and trusting the DUC?